

A third short table option is given by `pgfplotstableread`:

```
\pgfplotstableread[row sep=\\,col sep=&]{
M & Palermo & Milan \\
1 & 71.6 & 58.7 \\
.. & ... & ... \\
12 & 80.0 & 61.7 \\
}\dataRain
```

This meteorological data has been published in References 2 and 3. If the table is a bit longer, I can continue to use `pgfplotstableread`, but in another way, as shown below:

```
\pgfplotstableread{absorbance.txt}{\dataabs}
```

...in which `absorbance.txt` is a simple text file with a structure comprising rows and columns. In this example, `usepackage{pgfplotstable}` must be added in the preamble (see the next section). Probably, the simplest way is to put the data file, for example `pressure.txt`, directly after `addplot`, without any other command:

```
\addplot [p1] table[x=Date,y=DAP]{pressure.txt};
```

This last example will be better explained in the next section.

File structure

A generic LaTeX file has the following well-known structure:

```
\documentclass{...}
...
\begin{document}
...
\end{document}
```

...in which the preamble is the part between `documentclass` and `begin{document}`. The following example is a bit more detailed:

```
\documentclass[border=0.5cm]{standalone}
\usepackage[svgnames]{xcolor}
\usepackage{pgfplots}
\usetikzlibrary{pgfplots.dateplot}
\pgfplotsset{width=16cm,height=12cm}
\begin{document}
\begin{tikzpicture}
\begin{axis}[
date coordinates in=x,
grid=major,
grid style={dashed,gray!30},
xtick align=outside,
ytick align=outside,
```

```
xtick pos=left,
ytick pos=left,
xticklabel=\month.\year,
xticklabel style={rotate=90,anchor=near xticklabel},
ylabel={Pressure (mmHg)},
title={Blood pressure},
title style={font=\huge},
legend cell align=left,
p1/.style={draw=DodgerBlue,line width=1pt,mark=*,mark
options={fill=white}},
p2/.style={draw=Salmon,line width=1pt,mark=*,mark
options={fill=white}}
]
\addplot [p1] table[x=Date,y=DAP]{pressure.txt};
\addplot [p2] table[x=Date,y=SAP]{pressure.txt};
\legend{DAP,SAP}
\end{axis}
\end{tikzpicture}
\end{document}
```

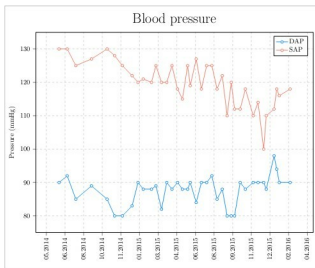


Figure 1: A time series

The preamble contains the packages, the libraries and some settings to build the plot — typically, these are the width, height and, eventually, the definition of a custom colour map. In this example, the `xcolor` package is used to have `svgnames` for the colours — for example, `DodgerBlue`, `Salmon`, etc (see Reference 4, pages 38–39). Usually, the preamble also contains the data table if it's used with `pgfplotstableread`. The part between `begin{axis}` and `end{axis}` is about the plot options — for example, the grid, ticks, labels, title and legend. In this part, `addplot` is also present. It's used to add each data series to the plot. Each data series has its own style options as specified with `p1` and `p2`. The file `pressure.txt` contains a simulation of medical data about the measure of the diastolic (DAP) and systolic (SAP) arterial pressures during a certain period of time.